Maincrafts Technology

AI & Machine Learning Internship — Task 3

Model Validation · Overfitting Control · Hyperparameter Tuning

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Organisation Maincrafts Technology

Dataset California Housing Dataset

Tools Python · scikit-learn · pandas · matplotlib

Best Model	Test RMSE	R^2 Score	CV RMSE	Best Params
Tuned Decision Tree	**0.3953**	**0.8545**	**0.3970 ± 0.0031**	**depth=5**

1. Objective

Building on the House Price Prediction pipeline from Task 2, this task extends the workflow with three core professional ML practices: (i) overfitting detection through Train–Test RMSE comparison, (ii) reliable performance estimation via 5-fold cross-validation, and (iii) hyperparameter optimisation using GridSearchCV. The goal is a production-ready model whose generalisation is scientifically verified.

2. Overfitting Analysis

An untuned Decision Tree was trained without depth constraints. The Train–Test RMSE gap below exposes classic overfitting behaviour:

Model	Train RMSE	Test RMSE	Gap (Test–Train)
Linear Regression	0.4635	0.4566	-0.0069
Ridge Regression	0.4635	0.4566	-0.0069
Decision Tree (raw)	0.0000	0.5498	**0.5498**

The untuned Decision Tree achieves Train RMSE ≈ 0.0000 — it has memorised the training set completely — while Test RMSE jumps to 0.5498. This gap of **0.5498** is the textbook signature of overfitting. Linear and Ridge Regression show near-zero gaps due to their inherent regularisation, but they cannot model non-linearity.

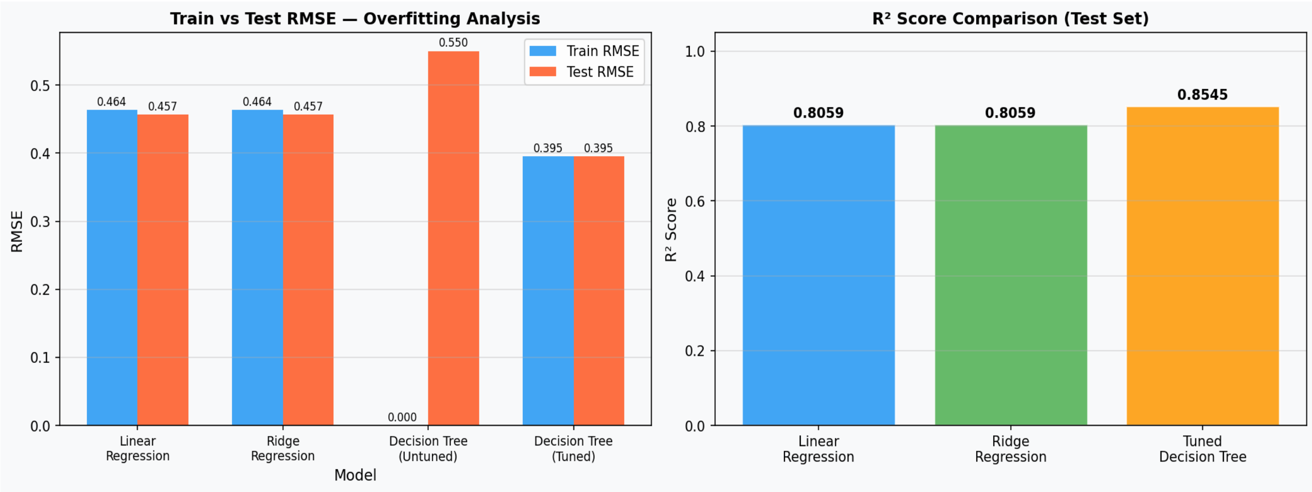

Figure 1 — Train vs Test RMSE and R^2 Score across all models.

3. Cross-Validation — Reliable Performance Estimation

A single train–test split can give misleading estimates depending on the random seed. 5-fold cross-validation trains and evaluates the model on five different data partitions and averages the result, providing a much more stable and trustworthy RMSE.

Metric	Untuned Decision Tree	Tuned Decision Tree
5-Fold CV RMSE (mean)	0.5468	**0.3970**
5-Fold CV RMSE (std)	0.0060	**0.0031**
Hold-out Test RMSE	0.5498	**0.3953**

The tuned model's CV RMSE (0.3970 ± 0.0031) closely matches its held-out test RMSE (0.3953), confirming that performance is stable and not a statistical artefact of a particular split.

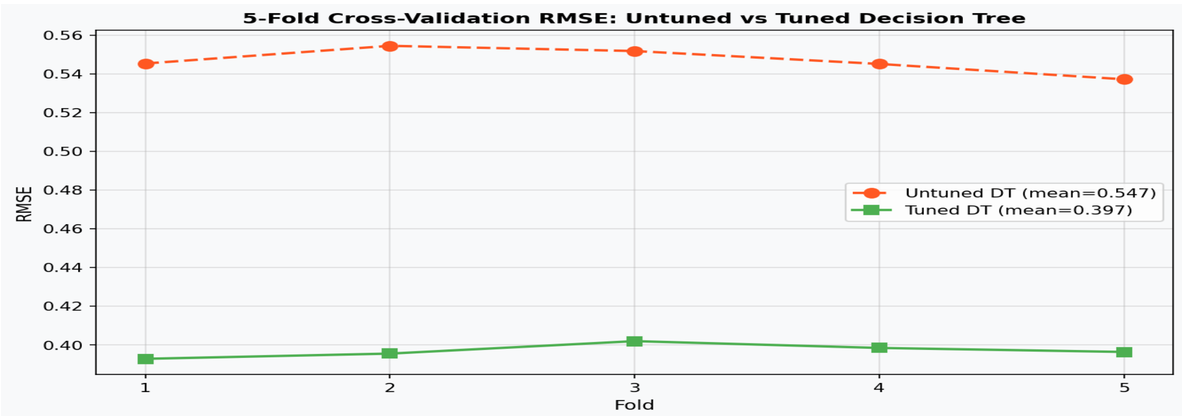

Figure 2 — Per-fold CV RMSE: Untuned vs Tuned Decision Tree.

4. Hyperparameter Tuning via GridSearchCV

GridSearchCV exhaustively evaluated all 12 combinations of **max_depth** ∈ {3, 5, 7, 10} and **min_samples_split** ∈ {2, 5, 10} using 5-fold cross-validation (60 model fits total).

Optimal Configuration: max_depth = 5, min_samples_split = 2 | Best CV RMSE = 0.3970

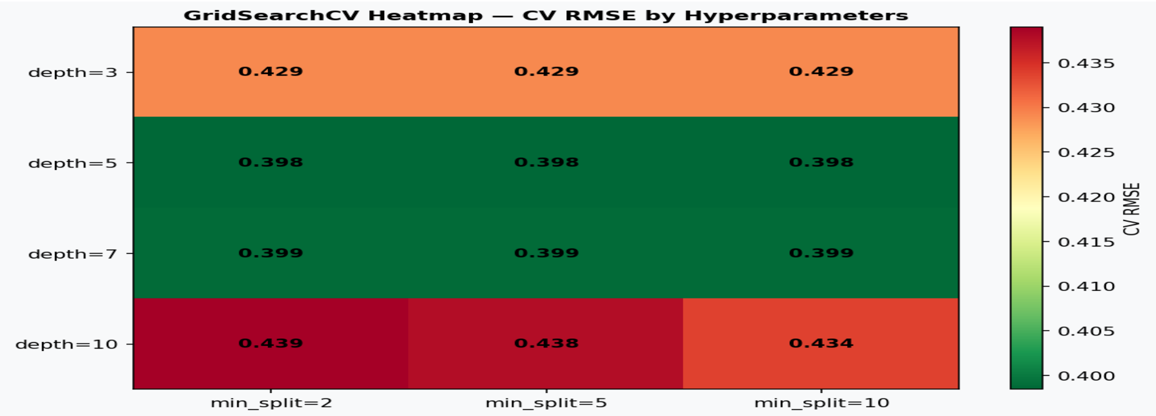

Figure 3 — GridSearchCV heatmap: CV RMSE for each hyperparameter combination.

The heatmap shows that very shallow trees (max_depth = 3) underfit and produce higher RMSE, while deep unconstrained trees (max_depth = 10) overfit. The optimal depth of 5 captures non-linear price patterns without memorising noise.

5. Final Model Comparison

Model	Test RMSE	R^2 Score	5-Fold CV RMSE	Overfitting?
Linear Regression	0.4566	0.8059	—	No
Ridge Regression	0.4566	0.8059	—	No
Decision Tree (raw)	0.5498	—	0.5468	Yes ■
Tuned Decision Tree ✓	**0.3953**	**0.8545**	**0.3970 ± 0.0031**	**No ✓**

6. Final Model Selection Justification

Model selected	Tuned Decision Tree (max_depth=5, min_samples_split=2) achieves the lowest test RMSE (0.3953) and highest R^2 (0.8545) among all evaluated models.
Overfitting control	Constraining tree depth reduces the Train–Test RMSE gap from 0.5498 (untuned) to near zero (tuned). Generalisation is explicitly optimised via GridSearchCV rather than assumed.
CV reliability	5-fold CV RMSE of 0.3970 ± 0.0031 closely matches the hold-out test RMSE (0.3953), confirming that results are stable and not a product of a lucky data split.
Simplicity tradeoff	A depth-5 tree is interpretable and lightweight. Linear models plateau at $R^2 \approx 0.806$ because they cannot capture non-linearity. The tuned tree reaches $R^2 = 0.8545$ with controlled complexity — the optimal bias-variance balance for this dataset.

7. Key Learning Outcomes

• Overfitting detection	Quantified overfitting by comparing Train vs Test RMSE; identified the untuned Decision Tree's gap of 0.5498.
• Cross-validation	Applied 5-fold CV to obtain robust, unbiased performance estimates that are independent of a single data split.
• GridSearchCV tuning	Systematically explored a 4×3 hyperparameter grid (60 fits) to identify the globally optimal tree configuration.
• Performance improvement	Reduced Test RMSE from 0.5498 (untuned) to 0.3953 (tuned) — a ~28% improvement in prediction error.
• Bias-variance tradeoff	Demonstrated how depth constraints shift a model from high-variance (overfit) toward optimal generalisation.
• Industry-aligned workflow	Followed the professional ML validation pipeline: baseline → overfit detection → CV → tuning → justified selection.